

● HARD

● INFORMATION AND IDEAS

● ANSWER KEY

Information and Ideas

17

10 answers · Rationale-rich · Teach from doc

Q1**B**

- B. although *A. mexicanus* is a single species at present, it could be in the process of splitting into distinct populations with different characteristics.

Choice B is the best answer because it presents the conclusion that most logically follows from the text's discussion of *A. mexicanus*. According to the text, *A. mexicanus*, a river-dwelling fish species, has colonized caves. The fish that live in rivers and those that live in caves show no significant genetic differences and are all capable of making the same sounds. The text indicates, however, that Hyacinthe and colleagues found that sounds that the river fish use in a particular context and for a particular purpose are used in a different context and for a different purpose by the cave fish. Additionally, the sounds made by cave fish show some differences depending on the specific cave the fish inhabit. The text goes on to state that Hyacinthe and colleagues have noted that differences in how the fish use sound to communicate could eventually become so great that they prevent fish from different locations from interbreeding. In other words, the river fish might eventually only breed with other river fish (with whom they share characteristics regarding sonic communication that they do not share with cave fish), while the cave fish might only breed with other cave fish for a similar reason. In context, this observation suggests that even though the fish are a single species right now, they could be in the process of splitting into distinct populations with different characteristics.

Choice A is incorrect because there is no information in the text suggesting that the river fish are less reliant on sonic communication than the cave fish are. Although the text does indicate that the river fish and cave fish are genetically similar, the text describes both groups as using sonic communication and says nothing to indicate that one group depends on that communication more than the other group does. Choice C is incorrect. The text states that all members of the species can emit the same sounds but that the function and context of sounds differ depending on whether the fish live in rivers or caves, but it does not indicate that river fish produce sounds that cave fish do not or vice versa. Choice D is incorrect because it contradicts the text. The text says that there is little genetic difference between the river fish and the cave fish, not that the river fish and cave fish are so genetically distinct that they can be considered separate species.

Q2

A

- A. the selection process that enabled the Navajo-Churro to retain its somewhat coarse outer fiber also resulted in inner fiber that, at its softest, is softer than either the mouflon's or the Merino's inner fiber.

Choice A is the best answer because it accurately uses data from the table to complete the statement about the fibers of three types of sheep. The text discusses some qualities of outer and inner fibers of mouflons, Spanish Merino sheep, and Navajo-Churro sheep and indicates that sheep fibers with finer, or smaller, diameters are softer than fibers with larger diameters. The text then sets up a conclusion that can be made by comparing the fibers of all three sheep types. The table shows that the minimum diameter of the Navajo-Churro's inner fibers is 10 microns, which is smaller than the minimum diameters of the inner fibers of both the mouflon (15 microns) and the Spanish Merino (17 microns). Thus, comparing the fiber characteristics of all three sheep reveals that at its softest, the inner fiber of the Navajo-Churro is softer than either the mouflon's or the Merino's inner fiber.

Choice B is incorrect because it doesn't accurately describe the data in the table. The table shows that the softness of Navajo-Churro inner fiber isn't uniform (since the diameter of Navajo-Churro inner fiber ranges from 10 microns to 35 microns) and that the diameter of the outer fiber of the mouflon isn't variable (rather, it is a uniform 150 microns). Choice C is incorrect because it doesn't accurately describe the data in the table. The text indicates that both Spanish Merino sheep and Navajo-Churro sheep were domesticated from mouflon sheep. The table doesn't show that domestication increased the minimum diameter of inner fibers for both Spanish Merino sheep and Navajo-Churro sheep from the 15-micron diameter of mouflon inner fibers: the minimum diameter of Spanish Merino increased to 17 microns, but the minimum diameter of Navajo-Churro inner fibers *decreased* to 10 microns. Choice D is incorrect because it doesn't accurately describe the data in the table. The text explains that sheep fibers with finer, or smaller, diameters are softer than fibers with larger diameters. The table does show that Spanish Merino outer fibers are finer, and thus softer, than mouflon outer fibers and that at their finest, Navajo-Churro outer fibers are softer than mouflon outer fibers, but it also shows that Spanish Merino inner fibers are less soft than mouflon inner fibers, since their minimum diameter is greater than 15 microns, and that Navajo-Churro inner fibers can be less soft than mouflon inner fibers, since the diameter ranges up to 35 microns.

Q3

B

- B. the reproductive success of *A. ocellaris* would be at risk if they were to selectively settle in regions that are regularly exposed to low levels of ALAN.
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Choice B is the best answer because it most logically completes the text's discussion of artificial light at night (ALAN) and the common clownfish (*A. ocellaris*). The text indicates that researchers found that when *A. ocellaris* is exposed to low levels of ALAN, spawning frequency and egg fertilization aren't significantly affected but hatching is completely inhibited—that is, the eggs never hatch. Therefore, if *A. ocellaris* were to settle only in regions where the fish would be regularly exposed to low levels of ALAN, their eggs would stop hatching and their reproductive success would be at risk.

Choice A is incorrect because the text indicates that exposure to low levels of ALAN had no significant effect on egg fertilization for *A. ocellaris*, so there's no reason to expect there would be any significant difference in rates of successful egg fertilization between areas with low levels of ALAN and areas without ALAN. Choice C is incorrect because the text doesn't discuss the particular effects of low levels of ALAN on any species of coral reef fish besides *A. ocellaris*. For this reason, there's no support in the text for the idea that the reproductive success of *A. ocellaris* is more greatly affected by the presence of low levels of ALAN than the reproductive success of other species of coral reef fish is. Choice D is incorrect. The text does indicate that *A. ocellaris* incubation was most strongly affected by low levels of ALAN, but it doesn't indicate that there was a greater effect on spawning frequency than on egg fertilization; in fact, the text states that there was no significant effect on either.

Q4

C

- C. The roots of *B. tomentosa* and *B. macrantha* carve new entry points into rocks even when cracks in the surface are readily available.
-

Choice C is the best answer because it presents a finding that, if true, would support the researchers' hypothesis about the plants' dependence on dissolving rock. The text indicates that the roots of the two plant species grow directly into quartzite rock, where hairs on the roots secrete acids that dissolve the rock. The researchers hypothesize that the plants depend on this process because dissolving rock opens spaces for the roots to grow and releases phosphates that provide the plants with phosphorus, a vital nutrient. If the plants carry out this process of dissolving rock even when the rock already has spaces into which the roots could grow, that would support the researchers' hypothesis because it suggests that the plants are getting some advantage—such as access to phosphorus—from the action of dissolving rock. If the plants don't benefit from dissolving rock, they would be expected to grow in the cracks that already exist, as doing so would mean that the plants don't have to spend energy creating and secreting acids; if, however, the plants create new entry points by dissolving rock even when cracks already exist, that would support the hypothesis that they depend on dissolving rock for some benefit.

Choice A is incorrect because the existence of soil-inhabiting members of the Velloziaceae family with similar root structures to those of the two species discussed in the text wouldn't support the researchers' hypothesis that the species discussed in the text depend on dissolving rock. If other such members exist, that might suggest that the root structures can serve more functions than secreting acids to dissolve rock (since dissolving rock may not be necessary for plants living in soil), but that wouldn't suggest anything about whether the species discussed in the text benefit from dissolving rock. Choice B is incorrect because differences in the proportions of citric and malic acid secreted by the two species would be irrelevant to the hypothesis that the plants depend on dissolving rock. There's no information in the text to suggest that the proportion of each acid has any bearing on the process of dissolving rock or on any benefits the plants might receive from that process. Choice D is incorrect because if the two species thrive on rocks without phosphates, that would weaken the researchers' hypothesis that the plants depend on dissolving rock partly because dissolving rock gives them access to phosphates. If the plants can survive on rocks without getting a vital nutrient by dissolving those rocks, then either the nutrient isn't actually vital for those plants or they can get the nutrient in some way other than by dissolving rocks.

Q5

D

- D. Blue icebergs and green icebergs contain similarly small traces of dissolved organic carbon.
-

Choice D is the best answer because it presents a finding that, if true, would weaken the scientists' hypothesis about icebergs that appear to be green. The text indicates that most icebergs are either mostly white or blue in color but that some icebergs in Antarctica appear to be green. The text goes on to say that the scientists hypothesized that this green color occurs when yellow-tinted dissolved organic carbon in ocean waters mixes with blue ice. A finding that both blue icebergs and green icebergs contain similarly small traces of dissolved organic carbon would suggest that something other than yellow-tinted organic carbon causes some icebergs' green color, since the blue icebergs that contain yellow-tinted organic carbon remained blue instead of turning green.

Choice A is incorrect because, according to the text, the scientists' hypothesis was that blue icebergs, not white ones, change color when their ice mixes with yellow-tinted dissolved organic carbon. A finding that white ice, because of its air bubbles, doesn't change color when it's mixed with dissolved organic carbon would therefore have no bearing on the scientists' hypothesis. Choice B is incorrect because the text focuses only on Antarctic icebergs that appear to be green. It doesn't indicate that icebergs in locations other than Antarctica have been found to have a green hue. A finding that dissolved organic carbon has a stronger yellow color in Antarctic waters than in other places would therefore have no bearing on the scientists' hypothesis that green color in icebergs in Antarctica is caused by yellow-tinted dissolved organic carbon mixing with blue ice. Choice C is incorrect because, according to the text, the scientists' hypothesis was that blue icebergs turn green when their ice mixes with yellow-tinted dissolved organic carbon in the sea around them. If that's correct, one would expect blue icebergs and green icebergs to be located at a distance from each other since all blue icebergs in an area where the waters contain yellow-tinted dissolved organic carbon would take on a green hue. A finding that blue icebergs and green icebergs are rarely found near each other would therefore strengthen, not weaken, the researchers' hypothesis.

Q6

A

- A. Isoguanine and purine were detected in both meteorite samples but not in the soil sample.
-

Choice A is the best answer. The researchers concluded that the meteorite's nucleobases weren't the result of soil contamination. Presence of nucleobases in the meteorite and not in soil provides evidence that those nucleobases likely didn't come from the soil.

Choice B is incorrect. This choice doesn't justify the conclusion. The researchers concluded that the meteorite's nucleobases weren't the result of soil contamination. If the nucleobases are present in both the soil and meteorite, then it's possible that these nucleobases came from the soil. Choice C is incorrect. This choice misreads the table. Purine was not detected in the soil sample. Choice D is incorrect. This choice misreads the table. Both isoguanine and hypoxanthine were detected in both Murchison meteorite samples.

Q7

C

- C. On average, participants who watched the panel correctly answered more questions about the issue than those who watched the debate or the single commentator did.
-

Choice C is the best answer because it presents the finding that, if true, would most strongly support Tannen's hypothesis. According to the text, Tannen's hypothesis is that multiple perspectives presented in a noncompetitive format is more informative than a debate between opposing viewpoints is. If participants who saw a panel of three commentators with various views about an issue answered more questions about the issue correctly than did participants who saw a debate, that would support Tannen's hypothesis since it would show that participants who heard multiple varied perspectives were better informed than were participants who heard a debate between opposing viewpoints.

Choice A is incorrect because finding that participants perceived commentators in the debate as more knowledgeable than commentators in the panel is irrelevant to Tannen's hypothesis, which is that presenting multiple perspectives on an issue is more informative to the audience than presenting opposing views of the issue is. Participants' perception of how knowledgeable panelists are has no bearing on how much participants learn from the panelists. Choice B is incorrect because finding that participants perceived commentators in the panel as more knowledgeable than a single commentator is irrelevant to Tannen's hypothesis, which is that presenting multiple perspectives on an issue is more informative to the audience than presenting opposing views of the issue is. Participants' perception of how knowledgeable panelists are has no bearing on how much participants learn from the panelists, and Tannen's hypothesis says nothing about how informative single commentators are. Choice D is incorrect because finding that participants who watched a single commentator answered more questions correctly than participants who watched the debate did wouldn't be relevant to Tannen's hypothesis, which is that hearing multiple varying perspectives is more informative than hearing a debate. Tannen's hypothesis says nothing about how informative single commentators are.

Q8

A

- A. For pin flowers, damage led to longer times per flower in all pollinator genera, whereas for thrum flowers, damage led to longer times per flower only in *Habropoda* and *Osmia*.
-

Choice A is the best answer because it describes data from the table that support the researchers' conclusion that the effect of floral damage on time spent per flower varied by both floral morph and the genus of the pollinator. The table presents the average time pollinators spent per floral morph. The data in the table shows that for pin flowers, the average time spent per flower by all pollinator genera was higher for flowers that had been artificially damaged than for intact flowers. By contrast, for thrum flowers, the difference in time spent on intact and damaged flowers is seen for only some pollinator genera. This supports the researchers' conclusion that both floral morph and the genus of the pollinator are factors involved in the effect that floral damage has on time pollinators spend per flower.

Choice B is incorrect. The table shows that pollinators belonging to the genus *Xylocopa* spent less time on both damaged pin flowers and damaged thrum flowers than pollinators belonging to the genus *Osmia* did. Choice C is incorrect. The table shows that the artificial damage to the thrum flowers led to shorter average times spent by pollinators on those flowers for only two of the four pollinator genera represented in the table, not three. Furthermore, this choice doesn't address the effect the artificial damage had on pin flowers; thus, even if accurate, this evidence doesn't support the researchers' conclusion that both floral morph and pollinator genus affect time spent per flower when a flower has been damaged. Choice D is incorrect. While this statement is true based on the information in the table, it doesn't provide evidence that supports the researchers' conclusion that the effect of floral damage on time spent per flower varied by both floral morph and the genus of the pollinator because this choice provides information only about how much time one pollinator genus spent on intact flowers.

Q9

B

- B. Fermented beans contained significantly less soluble fiber and raffinose than nonfermented beans, and when cooked, the fermented beans also displayed a significant reduction in trypsin inhibitors and tannins.
-

Choice B is the best answer because it presents a finding that would best support Granito and Álvarez's claim that fermenting black beans makes them easier to digest and more nutritious. The text indicates that high levels of soluble fiber and raffinose in black beans make the beans hard to digest and that tannins and trypsin inhibitors make it harder for the body to extract nutrients from the beans. If it were found that fermenting the beans significantly reduces their levels of soluble fiber, raffinose, trypsin inhibitors, and tannins when cooked, this would directly support the claim that fermentation improves the digestibility of the beans and makes them more nutritious.

Choice A is incorrect because the text indicates that trypsin inhibitors and tannins interfere with the body's ability to extract nutrients from black beans; if fermentation and cooking were found to increase these antinutrients, fermented beans would likely be less nutritious than unfermented ones, not more nutritious (as Granito and Álvarez claim). Choice C is incorrect because the text doesn't address the idea that greater nitrogen absorption in the gut has an effect on a food's digestibility or level of nutrition, so the discovery of the presence of microorganisms that may increase nitrogen absorption wouldn't provide relevant support for the claim that fermentation makes black beans easier to digest and more nutritious. Choice D is incorrect because Granito and Álvarez's claim focuses on the effect of fermenting black beans, but the finding that nonfermented black beans also have fewer trypsin inhibitors and tannins when cooked at high pressure would suggest that the role of the cooking method could be significant when it comes to nutrition; further, the finding wouldn't address the beans' digestibility.

Q10

C

C. *DptB* may have developed as a specific defense against *A. sicerae*.

Choice C is the best answer because it most logically completes the text based on supporting data in the graph. The text indicates that in the fly *D. melanogaster*, *DptA* and *DptB* are genes that encode peptides that both fight pathogens and promote beneficial microbes. Researchers tested *D. melanogaster*'s resistance to *P. rettgeri* and *A. sicerae* bacteria based on which variation of the peptide-encoding gene the flies exhibit: *DptA* silenced (referred to as type A), *DptB* silenced (type B), or both silenced (type AB). The text also indicates that resistance to *P. rettgeri* correlates with *DptA* activity but not with *DptB* activity (which would manifest as type B flies surviving at a higher rate than other fly types when exposed to *P. rettgeri*). The graph shows the post-*A. sicerae* infection results, which indicate that *DptB* activity was most strongly associated with survival, whereas *DptA* activity was not (manifesting in the graph as the type A flies having greater survival rates than the other fly types). In other words, when *DptA* activity was silenced, the flies showed relatively high survival rates, but when *DptB* activity was silenced, whether on its own or in conjunction with *DptA* activity being silenced, survival rates were low, suggesting that *DptB* may have developed as a specific defense against *A. sicerae*.

Choice A is incorrect. The graph suggests that *DptA* activity is associated with a low rate of survival, not a high one. Furthermore, the graph shows results for flies where *DptA* alone was silenced, *DptB* alone was silenced, and both were silenced and thus does not show any flies with activity in both *DptA* and *DptB*, which would be necessary to determine whether *DptA* conferred defense against *A. sicerae* in the presence of *DptB*. Choice B is incorrect. Only two bacteria species were considered in the text: *P. rettgeri* and *A. sicerae*. The text and graph taken together suggest that activity in *DptA* is associated with resistance to *P. rettgeri* while *DptB* activity is not, and that *DptB* activity is associated with resistance to *A. sicerae* while *DptA* is not. There is no further information to suggest one genetic type confers resistance to a greater number of pathogens than the other. Choice D is incorrect. The graph does not address flies with activity in both *DptA* and *DptB*. All flies represented in the graph had one or both of *DptA* and *DptB* silenced, or inactive.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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